

Exercise 6:

Draw a Use case diagram to model for a quiz system. A user can request a quiz for the system. The system picks a set of questions from its database, and composes them together to make a quiz. It rates the user's answers and gives hints if the user requests it. In addition to users, we also have helpers who provide questions and hints. And also, administrators who must certify questions to make sure they are not too trivial, and that they are correct

Aim:

To design a UML Use Case Diagram for a Quiz System that models the interactions between users, helpers, and administrators in managing and taking quizzes.

Procedure:

Step 1: Identify Key Actors

User

Requests a quiz

Answers quiz questions

Requests hints

Receives ratings

Helper

Provides questions

Provides hints

Administrator

Certifies questions (ensures correctness and difficulty level)

Quiz System

Selects questions from the database

Composes quizzes

Evaluates user answers

Rates users

Step 2: Define Use Cases for Each Actor

User

Request Quiz

Answer Questions

Request Hint

Receive Rating

Helper

Provide Questions

Provide Hints

Administrator

Certify Questions

Quiz System (automated processes)

In the Use Case Diagram, actors interact with use cases as follows:

"Rate User"

The screenshot displays the Umbrello UML Modeler software interface. The main workspace shows a Use Case Diagram for a 'quiz_system.xmi' file. The diagram includes three actors: 'User', 'Helper', and 'Administrator'. A central box contains several use cases: 'review out', 'take quiz', 'request hint', 'review rating', 'provide question', 'provide hint', 'manage database', and 'certify question'. Lines connect the actors to their respective use cases: 'User' to 'review out', 'take quiz', 'request hint', and 'review rating'; 'Helper' to 'provide question' and 'provide hint'; and 'Administrator' to 'manage database' and 'certify question'. The left sidebar contains four panels: 'Stereotypes' (showing a table with 'Name' and 'Usage'), 'Tree View' (showing a hierarchical list of elements), 'Debug' (showing a list of classes), and 'Documentation' (showing a list of elements). The bottom status bar indicates the system is 'Ready' and shows various system icons and the time '09:23 AM 11-02-2023'.

Thus the UML diagram has been implemented successfully.

